



Part A – General HCI, Methods & Cognition (Questions 1-16)

Question No. 1 (3.7 pts.)

Which statement about heuristic evaluation is most accurate as a limitation relative to user testing?

- ☒ a. It can systematically identify guideline violations early, but may miss issues driven by domain misconceptions or novel user strategies
- ☐ b. It can under-represent workflow issues that emerge only across multi-step, multi-screen journeys
- ☐ c. It is sensitive to evaluator expertise and can vary in severity ratings across evaluators
- ☐ d. It can generate false positives when guidelines conflict with domain-specific conventions
- ☐ e. It may overlook context-of-use factors (interruptions, constraints) when performed offsite

Question No. 2 (3.7 pts.)

A dispatch interface is used during emergencies with constant interruptions, multitasking, and handoffs. Which method most directly targets discovering interruption-driven failures?

- ☐ a. An expert heuristic review focusing on consistency and error prevention
- ☐ b. A log review of task sequences to infer where delays or retries occur
- ☐ c. A short series of semi-structured interviews focusing on “hard moments” and typical workarounds
- ☐ d. A moderated usability test using realistic scenarios, conducted in a controlled quiet room
- ☒ e. Contextual inquiry/shadowing during real shifts to capture interruptions, handoffs, and workarounds

Question No. 3 (3.7 pts.)

Which scenario most strongly warrants prioritizing error prevention over recovery?

- ☐ a. Posting publicly where deletion exists but may not remove external copies
- ☐ b. Editing a document where Undo exists but users may not notice it under time pressure
- ☐ c. Approving a purchase where cancellation is possible within a short grace period
- ☐ d. Scheduling where changes are reversible but may trigger downstream notifications
- ☒ e. Confirming a high-stakes irreversible action with safety constraints (e.g., transferring funds to a new payee with no rollback)

Question No. 4 (3.7 pts.)

Users categorize “Requests,” “Cases,” and “Tickets” differently from the current menu structure. Which method best diagnoses category mismatch?

- ☐ a. Log/analytics review of navigation paths to infer where users deviate or backtrack
- ☐ b. Hybrid card sorting (open → propose structure; closed → validate fit) followed by synthesis
- ☐ c. Moderated usability testing on navigation tasks to see where users expect items to appear
- ☒ d. A/B testing two alternative navigation groupings to compare findability success rate
- ☐ e. Heuristic evaluation focusing on “match between system and the real world” and labeling consistency



**Question No. 5 (3.7 pts.)**

Why is it risky to rely only on clickstream analytics to identify “the usability problem” in a form abandonment funnel?

- ☒ a. Analytics shows where drop-off happens (what), but not the cognitive cause (why) without complementary qualitative methods
- b. Analytics can show correlations but rarely isolates a single causal UI element
- c. Confounds (device type, traffic source) can shift drop-off independent of UX
- ☒ d. Analytics may reflect mixed intents (comparison, postponing) rather than interface failure, requiring interpretation
- e. Instrumentation choices can bias what appears to be the “problem step”

Question No. 6 (3.7 pts.)

You can recruit only 6 target users. The key question is whether people interpret “Submit request” as final or reversible in a claims flow. Which approach best matches this goal?

- ☒ a. Heuristic evaluation focusing on error prevention and recovery to predict reversibility expectations
- ☒ b. Benchmarking time-on-task across users to infer whether the action feels final
- c. Card sorting to improve labeling of the claims categories so users feel in control
- ☒ d. A/B testing two “Submit” labels (e.g., “Submit” vs. “Send”) to compare completion rate
- ☐ e. Moderated usability testing with scripted critical tasks, observation, and a think-aloud protocol during task execution

Question No. 7 (3.7 pts.)

A team runs moderated sessions with target users on a clickable prototype and collects task success, time, critical errors, and behavior notes. Which classification fits best?

- ☒ a. Objective empirical evaluation
- b. Mixed empirical evaluation (objective metrics plus subjective ratings)
- c. Predictive evaluation (model-based performance estimation)
- ☒ d. Analytical expert-based evaluation (principle-based inspection)
- e. Subjective empirical evaluation (primarily perception-based)

Question No. 8 (3.7 pts.)

In testing, participants repeatedly miss a small but critical status indicator while focusing on a demanding multi-step form. What is the strongest explanation?

- a. Change blindness: users fail to notice updates when the indicator changes after an action
- b. Low visual salience: the indicator does not stand out relative to surrounding elements
- ☒ c. Expectation mismatch: users do not believe a status indicator should matter for task completion
- ☒ d. Banner blindness: users learn to ignore content in locations that resemble ads or promotional areas
- ☒ e. Inattentional blindness / attentional tunneling under cognitive load

**Question No. 9 (3.7 pts.)**

Users learn the system quickly in training, but after a few weeks they need to “re-learn” basic operations. Which usability attribute is most implicated?

- a. Perceived usability (satisfaction)
- b. Efficiency
- ☒ c. Memorability
- d. Error tolerance (recoverability)
- e. Learnability

Question No. 10 (3.7 pts.)

A team has storyboards and static screen flows for a government benefits portal. They must produce a prioritized list of usability risks in 48 hours, before any interactive prototype exists. What is the most defensible first step?

- a. Run a structured heuristic evaluation with multiple evaluators, then consolidate and severity-rank findings using agreed criteria
- ☒ b. Run a pluralistic walkthrough (designer + developer + stakeholder) to identify where explanations are needed
- ☒ c. Conduct 3–4 rapid moderated sessions using paper screens to estimate completion rates on end-to-end tasks
- ☒ d. Run a quick content audit and readability check to reduce cognitive burden in text-heavy screens
- ☒ e. Do stakeholder interviews to align on “what good looks like” before any evaluation activity

Question No. 11 (3.7 pts.)

During evaluation, the team wants to uncover users’ reasoning, misconceptions, and decision points while performing tasks (not just whether they succeed). Which method best fits this goal?

- ☒ a. Heuristic evaluation to identify guideline violations that may explain user confusion
- b. A/B testing two interface variants to compare completion and drop-off rates
- c. Analytics/log analysis to identify where users abandon the flow
- d. A post-session questionnaire (e.g., satisfaction/usability rating) to capture perceived difficulty
- ☒ e. Moderated usability testing using a think-aloud protocol during task execution

Question No. 12 (3.7 pts.)

A team proposes an A/B test to compare two onboarding concepts that differ in many ways (sequence, content, number of steps, and UI layout). At the same time, the team has not finalized a single primary success metric, and tracking is only partially implemented. What is the best critique?

- a. The main issue is only statistical power; once traffic is sufficient the conclusions will be causal
- ☒ b. The test would have weak causal interpretability because too many variables change at once, and outcomes are not reliably defined/measured yet
- c. A/B testing is appropriate here if the sample size is large enough, even without a primary metric
- d. A/B testing is better replaced by heuristic evaluation because it always yields clearer answers early
- e. The problem is only UI fidelity; with a high-fidelity prototype, A/B conclusions become reliable

**Question No. 13 (3.7 pts.)**

Two designs yield: Design X has higher task success but longer completion time; Design Y is faster but produces more critical errors. Which interpretation best distinguishes effectiveness and efficiency?

- ☒ a. The “better” design depends on context: risk tolerance, user expertise, and task criticality
- ☐ b. Y is more efficient and could be acceptable if critical errors are rare in real usage conditions
- ☒ c. X is more effective; Y may be more efficient on successful attempts, but at a cost to effectiveness due to critical errors
- ☐ d. X may reduce cognitive load but introduces friction; Y may encourage speed at the cost of accuracy
- ☐ e. The result reflects a speed-accuracy trade-off that should be evaluated with error severity, not time alone

Question No. 14 (3.7 pts.)

In a budgeting app, “Freeze Budget” uses a snowflake icon. Many users assume it means “hide temporarily” rather than “lock edits.” Which concept best explains the breakdown?

- ☒ a. Poor mapping between the control location and the object it affects within the layout
- ☒ b. Weak signifiers: the icon does not indicate whether it changes state or applies a constraint
- ☒ c. Overloaded terminology (“freeze”) that conflicts with other app features using similar wording
- ☒ d. A mental-model mismatch triggered by metaphor/semantics, leading to incorrect action expectations
- ☒ e. Insufficient feedback after the action, leaving users unsure what changed

Question No. 15 (3.7 pts.)

A stakeholder insists on high-fidelity prototypes because “users won’t take it seriously otherwise.” What is the best response grounded in prototyping trade-offs?

- ☐ a. Use high-fidelity for the main flow and low-fidelity for edge cases to balance realism and speed
- ☒ b. Use low-fidelity to iterate on structure quickly, then high-fidelity to validate interaction details once the core flow stabilizes
- ☐ c. Use high-fidelity primarily for stakeholder alignment, then validate later with users
- ☐ d. Use a medium-fidelity clickable wireframe to reduce bias while still supporting task execution
- ☐ e. Use two high-fidelity variants in parallel to accelerate learning through comparison

Question No. 16 (3.7 pts.)

You want fast expert feedback but worry experts will overweight rule violations that don’t matter in your domain. Which mitigation is best without losing speed?

- ☐ a. Use a pluralistic walkthrough including a domain practitioner to challenge assumptions
- ☐ b. Add a quick user satisfaction rating to prioritize expert findings
- ☐ c. Calibrate experts with a domain brief and a shared severity rubric before independent reviews
- ☒ d. Combine a brief heuristic pass with a small targeted usability test on the highest-risk domain-critical tasks
- ☐ e. Focus the expert review only on top tasks rather than the full UI



Part B - Scenario 1: Multimodal Ride-Hailing (Questions 17-20)

A ride-hailing app supports both voice and touch for "Request ride," "Cancel ride," "Share location," and "Safety check-in." Findings:

- Users are unsure whether the system executed a spoken command or merely recognized it.
- Accidental activation occurs for sensitive actions (especially cancellation).
- Under stress, users hesitate to use safety features and fear selecting the wrong option.

Question No. 17 (3.7 pts.)

Which accessibility improvement best complements voice interaction here?

- a. Provide adjustable speech rate and voice options for comprehension and comfort
- b. Offer an offline "low-connectivity" mode to reduce failure in constrained contexts
- ☒ c. Provide captions/transcripts of recognized speech and system responses for hearing-impaired users
- d. Allow personalized command synonyms and language/locale settings for broader inclusivity
- e. Provide a complete touch alternative with large targets, clear focus states, screen-reader labeling, and explicit feedback

***Question No. 18 (3.7 pts.)**

Which design change is most likely to reduce accidental activation of sensitive actions without blocking legitimate use?

- a. Require a wake word or explicit intent phrase before executing sensitive commands ("Confirm cancel ride")
- b. Add a short "grace period" allowing Undo after cancellation, reducing harm from accidental triggers
- c. Add multimodal feedback (haptic + visual) immediately, so users can stop before completion
- ☒ d. Context-appropriate confirmation (e.g., hold-to-confirm, two-step confirmation, or constrained confirmation under motion/noise)
- e. Use confidence-threshold logic: if ASR confidence is low, request clarification instead of executing

Question No. 19 (3.7 pts.)

Which principle most directly addresses uncertainty about whether a command executed?

- a. Match between system and real world by echoing the user's words in natural language
- b. Recognition rather than recall by keeping action options visible at all times
- c. Error prevention by reducing available actions during motion/noise
- d. Consistency by ensuring voice and touch use identical phrasing and iconography
- ☒ e. Visibility of system status through immediate, unambiguous feedback about outcomes and current state

***Question No. 20 (3.7 pts.)**

Which measurement approach best captures trust and perceived safety in high-stress interactions?

- a. Incident reports and support tickets labeled "safety concern" as an organizational signal
- b. Task-anchored self-report (confidence/trust ratings right after safety/cancel tasks) combined with observed hesitation markers
- c. A standardized post-session questionnaire (trust scale) not tied to specific tasks
- d. Longitudinal proxy: repeat usage of safety features over weeks
- e. Behavioral proxies only: time-to-act, abandonment, and error rates in safety flows



Part C - Scenario 2: Hospital Intranet Redesign (Questions 21-24)

A hospital relaunches an internal portal for physicians, nurses, and residents. Content is accurate, but staff struggle to find procedures, forms, on-call schedules, and safety protocols. The team plans to use personas, scenarios, and journey maps.

Question No. 21 (3.7 pts.)

When creating personas responsibly, which step typically consumes the most effort and why?

- ☒ a. Data collection + synthesis, because clustering patterns into stable archetypes requires triangulation and validation
- b. Validation sessions with staff to confirm personas are recognizable and actionable
- c. Stakeholder alignment workshops, because cross-department agreement takes time
- d. Writing persona narratives and goals, to avoid stereotypes and ensure precision
- e. Translating personas into requirements artifacts (scenarios, journey maps, KPIs)

Question No. 22 (3.7 pts.)

Which success criterion most directly validates whether the new portal structure is working?

- ☒ a. Faster page-load performance and fewer timeouts
- b. Reliable findability for critical items (task success with acceptable time and low critical error)
- ☒ c. Higher satisfaction scores in a post-launch "ease of use" survey
- d. Improved search success rate and fewer query refinements
- e. Reduced average clicks to reach high-traffic pages across all roles

Question No. 23 (3.7 pts.)

What is the most accurate difference between a journey map and a user scenario here?

- a. Journey maps align stakeholders end-to-end; scenarios help validate task flows and requirements
- b. A journey map is broader; a scenario is narrower and more testable as a script
- c. A journey map emphasizes emotions across phases; a scenario emphasizes concrete steps in one task
- ☒ d. A journey map can include multiple actors/systems; a scenario typically centers one persona and one goal
- e. A journey map spans stages and touchpoints over time; a scenario is a focused narrative of accomplishing a goal in a specific situation

Question No. 24 (3.7 pts.)

Given the problem statement, what domain should lead the intervention first?

- a. Content strategy (rewriting titles, reducing duplication) as the primary driver
- b. Search experience design (query support, relevance, filters) as the primary driver
- c. Change management/training (job aids, onboarding) as the primary driver
- ☒ d. Information architecture (findability, labeling, navigation structure)
- e. Interaction design (streamlining workflows) as the primary driver



Part D - Scenario 3: City Emergency Alerts App (Questions 25-27)

A city launches an emergency app: real-time instructions, "I'm safe" check-in, assistance requests, and quick forms. Under pressure, users report hesitation and errors.

Question No. 25 (3.7 pts.)

The app previously let experienced users trigger actions using short command phrases and a compact shortcut panel. The redesign replaces this with visible menus and icon groups listing all actions. After the change, critical errors drop, but completion time rises slightly. Which explanation best accounts for the pattern?

- a. The system shifts from recall-based input to recognition-based input, trading speed for reliability מספר (קיוס) יחיד
- b. Increased visibility reduces mistakes, but added UI elements slow scanning and decision time
- ☒ c. Learnability improves, but experts lose efficiency because practiced sequences are interrupted
- ☒ d. Reduced cognitive load lowers errors, but extra selection/confirmation steps add time
- e. Recognition support increases and errors fall, but interaction overhead can rise relative to expert shortcuts, reducing efficiency

Question No. 26 (3.7 pts.)

In shelters, a redesigned door has a large flat push plate on one side and a small pull handle on the other. People repeatedly pull when they should push. What best explains the failure?

- ☒ a. Insufficient feedback: the door provides no immediate response indicating correct action
- b. Missing constraints: the design does not make the wrong action harder than the right one
- c. Poor mapping: the relationship between action and response is unclear until after trying
- d. Weak signifiers: users do not notice the push plate as the interaction cue
- ☒ e. A mismatch between perceived affordance and required action, amplified by stress and time pressure

Question No. 27 (3.7 pts.)

The team wants to measure effectiveness of completing critical tasks ("Check-in safe," "Request help," "Find instructions"). Which metric set best captures effectiveness?

- a. Step drop-off rate + retries as an operational proxy
- b. Task completion rate + error rate (or critical error count), optionally with correctness checks
- c. Task completion rate + time-on-task to capture both success and speed
- d. NASA-TLX workload + post-task confidence ratings to capture perceived effectiveness
- e. First-click accuracy + time to first meaningful action as a proxy for correct intent

